

Lesson 5: Security Risks Part 1

45 minutes

Overview

Students investigate three different common security risks (phishing, keylogging, malware) in a jigsaw activity. In groups, students create Public Service Announcement slides warning of the dangers of their assigned security risk. Then students are grouped with students who investigated other security risks and are instructed to share their slide and give a voice over. The activity ends with the class coming together to discuss the security risks as a whole.

Standards

Full Course Alignment

CSP Conceptual Framework

- **IOC-2** - The use of computing innovations may involve risks to your personal safety and identity.

CSTA K-12 Computer Science Standards (2017)

- **NI** - Networks & the Internet

Agenda

Warm Up (5 minutes)

Activity (35 minutes)

Security Risks Jigsaw

Wrap Up (5 minutes)

Assessment: Check For Understanding

Objectives

Students will be able to:

- Discuss the warning signals for these common security risks
- Explain how these common security risks target people
- Identify common security risks: phishing, keylogging, malware, rogue access points

Preparation

- Read through the levels on Code Studio to familiarize yourself with the topics.
- Think through the logistics of running the jigsaw activity.
- Check the "**Teacher's Lounge**" forum for verified teachers to find additional strategies or resources shared by fellow teachers
- If you are teaching virtually, consider checking our **Virtual Lesson Modifications**

Links

Heads Up! Please make a copy of any documents you plan to share with students.

For the teachers

- **Security Risks Part 1** - Slides

▼ Make a Copy

Vocabulary

- **Keylogging** - the use of a program to record every keystroke made by a computer user in order to gain

fraudulent access to passwords and other confidential information

- **Malware** - software intended to damage a computing system or to take partial control over its operation
- **Phishing** - a technique that attempts to trick a user into providing personal information. That personal information can then be used to access sensitive online resources, such as bank accounts and emails
- **Rogue Access Point** - a wireless access point that gives unauthorized access to secure networks

Teaching Guide

Warm Up (5 minutes)

 **Discuss:** *Have you ever received an email or a text message that looked suspicious? Have you ever been unsure if you should open the message or click on a link? What are the things that made you suspicious?*

Discussion Goal In this discussion, we are previewing malicious links and common phishing attempts. If students do not have any of their own examples to offer up, try to be prepared with one of your own where you received an email that was clearly not wise to open.

Things that might make you suspicious:

- Misspellings of the person's name, email, or company
- Something that seems too good to be true ("Free Vacations for a year!")
- Asking for personal information
- A flashy link they want you to click on

Activity (35 minutes)

Security Risks Jigsaw

Remarks

There are many different ways that data can be stolen. Let's examine a few.

Group: Divide students into groups of two. Evenly divide the three topics among the groups:

1. Keylogging

 1

Keylogging

 Teaching Tip

If the keylogging articles linked in the lesson are blocked, here are a couple of alternate articles you can reference.

- [What is a Keylogger](#)
- [How to Prevent Keylogging Threats](#)

2. Phishing



Phishing

3. Malware



Malware

 **Do This (20 mins):** Students navigate to their assigned levels on Code Studio and examine their topic. After they have a good understanding of the content, students make a PSA (Public Service Announcement) slide covering the following things:

- What is the security risk?
- How are people targeted?
- What are the warnings?

 **Group (10 mins):** After students finish their slides, rearrange the class so there is a representative covering each topic in each group. Students share their slides with each other and give a voice over of the security risks.

 **Discuss (5 mins):** *Bring the class back together and ask for a volunteer from each of the topics to share their slide with the class and give a one minute overview.*

Teaching Tip

It's ok if the student who presents information has some inaccuracies in their reporting. Use this opportunity to correct misunderstandings and get everyone on the same page.

One thing that can come out of the discussion is the realization that Keylogging is a form of Malware. Malware is as broad category of malicious software that can collect information or exploit a system in many different ways.

Remarks

Another security risk that you may have heard about is a Rogue Access Point. How this works can get pretty complicated, but it's enough to know that a rogue access point is a wireless access point that gives unauthorized access to secure networks. This can be a physical device that is attached to a router - sometimes hidden from site! It can be detected in various ways, including looking for strange wireless signals.

Wrap Up (5 minutes)

Remarks

There are many different ways that you may be targeted to reveal sensitive information. We generally think of emails as being a safe way to communicate, but unsolicited emails, attachments, links and forms can all be used to compromise the safety and security of a computing system. These could come from people you don't know, or

from your friends and family who's security has been compromised.

It can be alarming to realize that there are many ways you are being targeted to reveal sensitive information. However, knowledge of the facts can help us be wiser consumers of technology. Later on in this unit we will explore further how to protect ourselves from these security risks.

 **Journal:** Record in your journal the following vocabulary words: Phishing, Keylogging, Malware, Rogue Access Point.

Assessment: Check For Understanding

Check For Understanding Question(s) and solutions can be found in each lesson on Code Studio. These questions can be used for an exit ticket.

Question: How would you explain these three security risks (phishing, keylogging, malware) to a family member? What would you say to help them understand the dangers?



Check For Understanding



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